

Superstore Sales Analysis (SQL Project)

Name: Payal Vijay Rananaware

Week: Week 3

Tools Used: MySQL Workbench

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-- PROJECT: SUPERSTORE SALES ANALYSIS (SQL)

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-- Objective:

-- To analyze Superstore dataset using SQL by performing data cleaning, transformation, and advanced queries.

-- Concepts Used:

-- Subqueries

-- Common Table Expressions (CTE)

-- Window Functions

-- Joins

-- Aggregations

-- Primary & Foreign Key Constraints

-- Dataset: Superstore Raw Dataset

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-- Step 1: Setup Data

CREATE DATABASE superstore_raw_database;

USE superstore_raw_database;

SELECT * FROM superstore_raw;

Result Grid		Filter Rows:		Export:	Wrap Cell Contents:		Fetch rows:			
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
▶	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles
	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale
	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale
	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord
	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle
	15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Off...	United States	Fort Worth
	16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Off...	United States	Fort Worth
	17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison
	18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan
	19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco
	20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco
	21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco

superstore_raw 39 x

Result Grid												Filter Rows:	Export:	Wrap Cell Contents:	Fetch rows:
	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit				
▶	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136				
	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.94	3	0	219.582				
	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.62	2	0	6.8714				
	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031				
	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164				
	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acces...	48.86	7	0	14.1694				
	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656				
	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152				
	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...	18.504	3	0.2	5.7825				
	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47				
	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092				
	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568				
	North C...	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967	15.552	3	0.2	5.4432				
	Washing...	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2	132.5922				
	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner...	68.81	5	0.8	-123.858				
	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders	2.544	3	0.8	-3.816				
	Wisconsin	53711	Central	OFF-ST-10004186	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 3...	665.88	6	0	13.3176				
	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer	55.5	2	0	9.99				
	California	94109	West	OFF-AR-10003056	Office Supplies	Art	Newell 341	8.56	2	0	2.4824				
	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone	213.48	3	0.2	16.011				
	California	94109	West	OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"	22.72	4	0.2	7.384				

DESCRIBE superstore_raw;

	Field	Type	Null	Key	Default	Extra
►	Row ID	int	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	text	YES		NULL	
	Ship Date	text	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	int	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

-- Creating normalized tables for Customers, Products, and Orders to structure the raw dataset into relational format

```
CREATE TABLE CUSTOMERS(  
    CUSTOMER_ID VARCHAR(50),  
    CUSTOMER_NAME VARCHAR(100),  
    SEGMENT VARCHAR(50),  
    REGION VARCHAR(50)  
);
```

```
CREATE TABLE PRODUCTS(  
    PRODUCT_ID VARCHAR(50),  
    CATEGORY VARCHAR(50),  
    SUB_CATEGORY VARCHAR(50),  
    PRODUCT_NAME VARCHAR(200)  
);
```

```
CREATE TABLE ORDERS(  
    ORDER_ID VARCHAR(50),  
    ORDER_DATE DATE,  
    CUSTOMER_ID VARCHAR (50),  
    PRODUCT_ID VARCHAR(50),  
    SALES DOUBLE,  
    QUANTITY INT,  
    PROFIT DOUBLE  
);
```

-- Inserting distinct records into above tables to avoid duplicates and maintain data consistency

```
INSERT INTO customers
```

```
SELECT DISTINCT
```

```
    `Customer ID`,
```

```
    `Customer Name`,
```

```
    `Segment`,
```

```
    `Region`
```

```
FROM superstore_raw;
```

```
INSERT INTO products
```

```
SELECT DISTINCT
    `Product ID`,
    `Category`,
    `Sub-Category`,
    `Product Name`
FROM superstore_raw;
```

```
INSERT INTO orders
SELECT
    `Order ID`,
    STR_TO_DATE(`Order Date`, '%m/%d/%Y'),
    `Customer ID`,
    `Product ID`,
    `Sales`,
    `Quantity`,
    `Profit`
FROM superstore_raw;
```

```
SELECT * FROM CUSTOMERS;
```

Result Grid				
		Filter Rows:		Export: 
		Wrap Cell Content: 		Fetch rows:
	CUSTOMER_ID	CUSTOMER_NAME	SEGMENT	REGION
▶	CG-12520	Claire Gute	Consumer	South
	DV-13045	Darrin Van Huff	Corporate	West
	SO-20335	Sean O'Donnell	Consumer	South
	BH-11710	Brosina Hoffman	Consumer	West
	AA-10480	Andrew Allen	Consumer	South
	IM-15070	Irene Maddox	Consumer	West
	HP-14815	Harold Pawlan	Home Office	Central
	PK-19075	Pete Kriz	Consumer	Central
	AG-10270	Alejandro Grove	Consumer	West
	ZD-21925	Zuschuss Donatelli	Consumer	West
	KB-16585	Ken Black	Corporate	Central
	SF-20065	Sandra Flanagan	Consumer	East
	EB-13870	Emily Burns	Consumer	West
	EH-13945	Eric Hoffmann	Consumer	West
	TB-21520	Tracy Blumstein	Consumer	East
	MA-17560	Matt Abelman	Home Office	Central
	GH-14485	Gene Hale	Corporate	Central
	SN-20710	Steve Nguyen	Home Office	Central
	LC-16930	Linda Cazamias	Corporate	Central
	RA-19885	Ruben Ausman	Corporate	West
	ES-14080	Erin Smith	Corporate	South
	ON-18715	Odella Nelson	Corporate	Central
	PO-18865	Patrick O'Donnell	Consumer	Central

SELECT * FROM PRODUCTS;

Result Grid				
Filter Rows:		Export:	Wrap Cell Content:	Fetch rows:
PRODUCT_ID	CATEGORY	SUB_CATEGORY	PRODUCT_NAME	
FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	
FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	
OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	
FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	
OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	
FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acces...	
OFF-AR-10002833	Office Supplies	Art	Newell 322	
TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	
OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...	
OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	
FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	
TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	
OFF-PA-10002365	Office Supplies	Paper	Xerox 1967	
OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine	
OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner...	
OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders	
OFF-ST-10004186	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 3...	
OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer	
OFF-AR-10003056	Office Supplies	Art	Newell 341	
TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone	
OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"	
OFF-AR-10000246	Office Supplies	Art	Newell 318	
OFF-AP-10001492	Office Supplies	Appliances	Acco Six-Outlet Power Strip. 4' Cord Length	

SELECT * FROM ORDERS;

Result Grid								
Filter Rows:								
Edit: Export/Import: Wrap Cell Content: Fets								
	ORDER_ID	ORDER_DATE	CUSTOMER_ID	PRODUCT_ID	SALES	QUANTITY	PROFIT	ROW_ID
▶	US-2014-106992	2014-09-19	SB-20290	TEC-MA-10003353	2519.958	7	-251.9958	1
	CA-2016-159695	2016-04-05	GM-14455	OFF-ST-10003442	158.368	7	13.8572	2
	CA-2016-115756	2016-09-05	PK-19075	OFF-ST-10000060	194.94	3	23.3928	3
	CA-2016-125318	2016-06-06	RC-19825	TEC-PH-10001433	328.224	4	28.7196	4
	CA-2016-109806	2016-09-17	JS-15685	OFF-AR-10004930	20.1	3	6.633	5
	CA-2016-152156	2016-11-08	CG-12520	FUR-BO-10001798	261.96	2	41.9136	6
	CA-2015-155040	2015-11-10	AH-10210	TEC-AC-10004469	79.9	2	35.156	7
	CA-2017-136826	2017-06-16	CB-12535	OFF-AR-10003602	14.016	3	4.7304	8
	CA-2016-115756	2016-09-05	PK-19075	OFF-ST-10003058	70.95	3	20.5755	9
	CA-2016-115756	2016-09-05	PK-19075	FUR-CH-10002372	242.94	3	29.1528	10
	CA-2016-111010	2016-01-22	PG-18895	OFF-FA-10003472	7.56	6	0.3024	11
	CA-2016-152156	2016-11-08	CG-12520	FUR-CH-10000454	731.94	3	219.582	12
	CA-2016-115756	2016-09-05	PK-19075	OFF-LA-10001317	22.05	7	10.584	13
	CA-2016-109806	2016-09-17	JS-15685	TEC-PH-10004093	73.584	2	8.2782	14
	CA-2016-109806	2016-09-17	JS-15685	OFF-PA-10000304	6.48	1	3.1104	15
	US-2017-145366	2017-12-09	CA-12310	OFF-ST-10004180	37.208	1	-7.4416	16
	CA-2017-154214	2017-03-20	TB-21595	FUR-FU-10000206	2.91	1	1.3677	17
	US-2017-145366	2017-12-09	CA-12310	OFF-EN-10004386	57.576	3	21.591	18
	CA-2017-163979	2017-12-28	KH-16690	OFF-ST-10003208	725.84	4	210.4936	19
	CA-2015-155334	2015-07-30	HA-14920	TEC-AC-10003628	209.93	7	92.3692	20
	CA-2015-155334	2015-07-30	HA-14920	FUR-FU-10003274	5.28	3	2.3232	21
	CA-2015-149587	2015-01-31	KB-16315	OFF-PA-10003177	12.96	2	6.2208	22
	CA-2016-166674	2016-04-01	RB-19360	OFF-AR-10000588	59.52	3	15.4752	23

SELECT COUNT(*)

FROM customers;

Result Grid	
	COUNT(*)
▶	2487

SELECT COUNT(*)

FROM products;

Result Grid	
	COUNT(*)
▶	1842

SELECT COUNT(*)

FROM orders;

Result Grid	
	COUNT(*)
▶	9694

-- Verifying successful data load and understanding dataset size

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-- STEP 2: ANALYTICAL SQL QUERIES

-- =====

-- Performing various business analysis using subqueries, CTEs, and window functions

-- 1) Find all orders where sales are greater than the average sales. (Subquery)

SELECT *

FROM ORDERS

WHERE SALES > (SELECT AVG(SALES)

FROM ORDERS);

Result Grid								
Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:		Fetch rows:
ORDER_ID	ORDER_DATE	CUSTOMER_ID	PRODUCT_ID	SALES	QUANTITY	PROFIT	ROW_ID	
US-2014-106992	2014-09-19	SB-20290	TEC-MA-10003353	2519.958	7	-251.9958	1	
CA-2016-125318	2016-06-06	RC-19825	TEC-PH-10001433	328.224	4	28.7196	4	
CA-2016-152156	2016-11-08	CG-12520	FUR-BO-10001798	261.96	2	41.9136	6	
CA-2016-115756	2016-09-05	PK-19075	FUR-CH-10002372	242.94	3	29.1528	10	
CA-2016-152156	2016-11-08	CG-12520	FUR-CH-10000454	731.94	3	219.582	12	
CA-2017-163979	2017-12-28	KH-16690	OFF-ST-10003208	725.84	4	210.4936	19	
US-2015-108966	2015-10-11	SO-20335	FUR-TA-10000577	957.5775	5	-383.031	28	
CA-2016-166674	2016-04-01	RB-19360	OFF-AR-10001953	263.88	6	71.2476	32	
CA-2017-147277	2017-10-20	EB-13705	FUR-TA-10001539	284.364	2	-75.8304	44	
CA-2017-147277	2017-10-20	EB-13705	OFF-ST-10000142	665.408	2	66.5408	45	
CA-2014-115812	2014-06-09	BH-11710	TEC-PH-10002275	907.152	6	90.7152	48	
CA-2014-115812	2014-06-09	BH-11710	FUR-TA-10001539	1706.184	9	85.3092	55	
CA-2014-115812	2014-06-09	BH-11710	TEC-PH-10002033	911.424	4	68.3568	56	
CA-2016-161389	2016-12-05	IM-15070	OFF-BI-10003656	407.976	3	132.5922	59	
US-2015-161991	2015-09-26	SC-20725	TEC-PH-10001760	1114.4	7	376.11	61	
CA-2015-130890	2015-11-02	JO-15280	FUR-TA-10002903	1038.84	5	51.942	62	
CA-2015-130883	2015-09-26	TB-21520	TEC-AC-10001956	239.8	5	47.96	64	
CA-2016-112697	2016-12-18	AH-10195	OFF-BI-10000778	254.058	7	-169.372	66	
CA-2016-112697	2016-12-18	AH-10195	OFF-SU-10000646	961.48	5	-204.3145	69	
CA-2016-110772	2016-11-20	NZ-18565	TEC-AC-10002001	255.984	2	54.3966	72	
CA-2014-111451	2014-12-26	KL-16555	FUR-FU-10004091	300.416	8	78.8592	74	
US-2016-157945	2016-09-26	NF-18385	FUR-CH-10002331	747.558	3	-96.1146	82	
CA-2015-109638	2015-12-15	JH-15985	TEC-AC-10004571	899.91	9	377.9622	86	

-- INSIGHT

-- A subset of orders contributes disproportionately higher revenue compared to the average order value.

-- 2) Find the highest sales order for each customer. (Subquery)

SELECT *

FROM ORDERS O

WHERE SALES = (SELECT MAX(SALES)

FROM ORDERS

WHERE CUSTOMER_ID = O.CUSTOMER_ID

);

Result Grid								
Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:		
ORDER_ID	ORDER_DATE	CUSTOMER_ID	PRODUCT_ID	SALES	QUANTITY	PROFIT	ROW_ID	
CA-2016-125318	2016-06-06	RC-19825	TEC-PH-10001433	328.224	4	28.7196	4	
CA-2016-152156	2016-11-08	CG-12520	FUR-CH-10000454	731.94	3	219.582	12	
US-2015-108966	2015-10-11	SO-20335	FUR-TA-10000577	957.5775	5	-383.031	28	
CA-2014-115812	2014-06-09	BH-11710	FUR-TA-10001539	1706.184	9	85.3092	55	
US-2015-161991	2015-09-26	SC-20725	TEC-PH-10001760	1114.4	7	376.11	61	
CA-2015-130890	2015-11-02	JO-15280	FUR-TA-10002903	1038.84	5	51.942	62	
CA-2016-112697	2016-12-18	AH-10195	OFF-SU-10000646	961.48	5	-204.3145	69	
CA-2014-111003	2014-06-01	CR-12625	OFF-AR-10002135	289.2	6	83.868	92	
CA-2016-109869	2016-04-22	TN-21040	FUR-TA-10001889	1272.63	6	-814.4832	100	
US-2015-101399	2015-01-17	JS-15940	FUR-FU-10002918	254.744	7	-312.0614	108	
CA-2015-129476	2015-10-15	PA-19060	TEC-AC-10000844	339.96	5	67.992	114	
CA-2015-106320	2015-09-25	EB-13870	FUR-TA-10000577	1044.63	3	240.2649	125	
US-2015-150630	2015-09-17	TB-21520	FUR-BO-10004834	3083.43	7	-1665.0522	139	
CA-2016-100790	2016-06-26	JG-15805	OFF-ST-10000689	704.25	5	84.51	146	
CA-2014-164973	2014-11-04	NM-18445	TEC-MA-10002927	3991.98	2	1995.99	148	
CA-2014-127691	2014-07-22	EM-14065	TEC-AC-10002567	159.98	2	57.5928	156	
CA-2017-140963	2017-06-10	MT-18070	FUR-BO-10001337	514.165	5	-30.245	161	
CA-2016-117590	2016-12-08	GH-14485	TEC-PH-10004977	1097.544	7	123.4737	173	
CA-2014-154627	2014-10-29	SA-20830	TEC-PH-10001363	2735.952	6	341.994	174	
CA-2015-117415	2015-12-27	SN-20710	FUR-BO-10002545	532.3992	3	-46.9764	180	
CA-2015-137946	2015-09-01	DB-13615	TEC-CO-10001449	959.984	2	335.9944	198	
CA-2014-129924	2014-07-12	AC-10420	FUR-TA-10004575	698.352	3	-17.4588	205	
CA-2014-113362	2014-09-14	AJ-10960	OFF-ST-10001809	449.15	5	8.983	216	

-- ALTERNATIVE APPROACH USING WINDOW FUNCTION

SELECT *

FROM (

SELECT

O.ORDER_ID,

O.CUSTOMER_ID,

O.SALES,

ROW_NUMBER() OVER (

PARTITION BY O.CUSTOMER_ID

ORDER BY O.SALES DESC

) AS RN

FROM ORDERS O

) T

WHERE RN = 1;

Result Grid					Filter Rows:	Export:
	ORDER_ID	CUSTOMER_ID	SALES	RN		
▶	CA-2016-103982	AA-10315	3930.072	1		
	CA-2016-131065	AA-10375	499.98	1		
	CA-2016-114601	AA-10480	479.97	1		
	CA-2015-110863	AA-10645	1323.9	1		
	CA-2016-140935	AB-10015	341.96	1		
	CA-2016-129714	AB-10060	4355.168	1		
	CA-2016-117121	AB-10105	9892.74	1		
	CA-2014-169061	AB-10150	479.97	1		
	CA-2015-115945	AB-10165	304.23	1		
	CA-2017-126662	AB-10255	479.984	1		
	US-2017-155425	AB-10600	899.136	1		
	CA-2014-129924	AC-10420	698.352	1		
	CA-2017-141789	AC-10450	1793.98	1		
	CA-2017-157987	AC-10615	1282.41	1		
	CA-2016-161928	AC-10660	201.584	1		
	CA-2017-127432	AD-10180	2999.95	1		
	CA-2016-146934	AF-10870	662.84	1		
	CA-2014-114181	AF-10885	349.965	1		
	CA-2016-105732	AG-10270	1336.44	1		
	CA-2014-147298	AG-10300	230.28	1		
	CA-2017-142909	AG-10330	182.55	1		
	CA-2014-158470	AG-10390	71.28	1		
	CA-2015-162201	AG-10495	516.96	1		

Result 49 x

-- Using ROW_NUMBER provides a more efficient and scalable way to identify highest value transactions per customer

-- INSIGHT

-- Each customer has a peak purchase behavior, useful for identifying high-value transactions.

-- 3) Calculate total sales for each customer. (CTE)

WITH CUSTOMER_SALES AS (

SELECT

CUSTOMER_ID,

```

SUM(SALES) AS TOTAL_SALES

FROM ORDERS

GROUP BY CUSTOMER_ID
)

SELECT * FROM CUSTOMER_SALES;

```

Result Grid			Filter Rows:	Export:	Wrap Cell
	CUSTOMER_ID	TOTAL_SALES			
▶	AA-10315	5563.56			
	AA-10375	1056.39			
	AA-10480	1782.872			
	AA-10645	4455.895			
	AB-10015	886.156			
	AB-10060	7732.669999999999			
	AB-10105	14355.610999999997			
	AB-10150	966.7099999999998			
	AB-10165	1113.838			
	AB-10255	898.084			
	AB-10600	1515.862			
	AC-10420	891.56			
	AC-10450	5527.8460000000005			
	AC-10615	2499.7560000000003			
	AC-10660	657.316			
	AD-10180	6100.999999999999			
	AF-10870	2052.9080000000004			
	AF-10885	861.5649999999999			
	AG-10270	2565.2579999999994			
	AG-10300	349.26			
	AG-10330	660.972			
	AG-10390	200.945			
	AG-10495	2338.679			

-- INSIGHT

-- Customer contribution to total revenue can be clearly measured.

-- 4) Find customers whose total sales are above average. (CTE + Subquery)

WITH CUSTOMER_SALES AS(

SELECT

CUSTOMER_ID,

SUM(SALES) AS TOTAL_SALES

FROM ORDERS

GROUP BY CUSTOMER_ID

)

SELECT *

FROM CUSTOMER_SALES

WHERE TOTAL_SALES > (

SELECT AVG(TOTAL_SALES)

FROM CUSTOMER_SALES

);

Result Grid			Filter Rows:	Export:
	CUSTOMER_ID	TOTAL_SALES		
▶	AA-10315	5563.56		
	AA-10645	4455.895		
	AB-10060	7732.669999999999		
	AB-10105	14355.610999999997		
	AC-10450	5527.8460000000005		
	AD-10180	6100.999999999999		
	AG-10675	3489.0395999999996		
	AG-10900	4510.7970000000005		
	AH-10075	3250.3369999999995		
	AH-10210	4805.344		
	AH-10690	7888.2940000000001		
	AI-10855	4375.785999999999		
	AJ-10780	4867.344		
	AJ-10795	4391.491		
	AM-10705	5653.861999999999		
	AP-10915	3323.556		
	AR-10540	6608.448		
	AR-10825	4523.335999999999		
	AS-10045	3035.0320000000006		
	AS-10090	3255.309999999999		
	AS-10225	4249		
	AS-10285	3212.272		
	AT-10735	3831.8640000000005		

Result 51 ×

-- INSIGHT

-- Only a small percentage of customers generate above-average revenue.

-- 5) Rank all customers based on total sales. (Window Function)

SELECT

CUSTOMER_ID,

SUM(SALES) AS TOTAL_SALES,

RANK() OVER(ORDER BY SUM(SALES) DESC) AS RANK_

FROM ORDERS

GROUP BY CUSTOMER_ID;

Result Grid				Filter Rows:	Export:
	CUSTOMER_ID	TOTAL_SALES	RANK_		
▶	SM-20320	25043.05	1		
	TC-20980	19017.847999999998	2		
	RB-19360	15117.339	3		
	TA-21385	14595.62	4		
	AB-10105	14355.610999999997	5		
	SC-20095	14142.333999999999	6		
	KL-16645	14071.917	7		
	HL-15040	12873.297999999999	8		
	SE-20110	12209.438	9		
	CC-12370	12129.072	10		
	TS-21370	11885.871	11		
	GT-14710	11820.119999999997	12		
	BM-11140	11609.900000000001	13		
	SV-20365	11420.645999999997	14		
	CJ-12010	11079.742	15		
	CL-12565	10880.545999999998	16		
	ME-17320	10635.917999999998	17		
	KF-16285	10604.266	18		
	BS-11365	10501.652999999998	19		
	EH-13765	10005.980000000001	20		
	JL-15835	9403.002999999999	21		
	GT-14635	9351.211999999998	22		
	HW-14935	9300.253999999997	23		

Result 52 ×

-- INSIGHT

-- Top-ranked customers are key contributors to business revenue.

-- 6) Assign row numbers to each order within a customer. (Window Function + PARTITION BY)

SELECT

ORDER_ID,

CUSTOMER_ID,

SALES,

ROW_NUMBER() OVER(PARTITION BY CUSTOMER_ID

ORDER BY SALES DESC) AS ROW_NUM

FROM ORDERS;

Result Grid		 Filter Rows:	Export: 	
	ORDER_ID	CUSTOMER_ID	SALES	ROW_NUM
▶	CA-2016-103982	AA-10315	3930.072	1
	CA-2014-128055	AA-10315	673.568	2
	CA-2016-103982	AA-10315	431.976	3
	CA-2017-147039	AA-10315	362.94	4
	CA-2014-128055	AA-10315	52.98	5
	CA-2016-103982	AA-10315	41.72	6
	CA-2015-121391	AA-10315	26.96	7
	CA-2014-138100	AA-10315	14.94	8
	CA-2014-138100	AA-10315	14.56	9
	CA-2017-147039	AA-10315	11.54	10
	CA-2016-103982	AA-10315	2.304	11
	CA-2016-131065	AA-10375	499.98	1
	CA-2015-140921	AA-10375	149.97	2
	CA-2017-100230	AA-10375	116.98	3
	CA-2015-114503	AA-10375	84.96	4
	US-2017-169488	AA-10375	39.96	5
	CA-2014-130729	AA-10375	34.272	6
	CA-2015-140921	AA-10375	28.4	7
	CA-2017-100230	AA-10375	17.94	8
	US-2017-169488	AA-10375	16.9	9
	CA-2016-126613	AA-10375	16.768	10
	CA-2014-158064	AA-10375	16.52	11
	CA-2017-100230	AA-10375	14.952	12

Result 53 ×

-- INSIGHT

-- Helps identify top transactions per customer.

-- 7) Display top 3 customers based on total sales. (Window Function)

```

WITH RANKED AS (
    SELECT
        CUSTOMER_ID,
        SUM(SALES) AS TOTAL_SALES,
        RANK() OVER(ORDER BY SUM(SALES) DESC) SALES_RANK
    FROM ORDERS
    GROUP BY CUSTOMER_ID
)
SELECT *
FROM RANKED
WHERE SALES_RANK <= 3;

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 			
	CUSTOMER_ID	TOTAL_SALES	SALES_RANK
▶	SM-20320	25043.05	1
	TC-20980	19017.847999999998	2
	RB-19360	15117.339	3

```

-- INSIGHT
-- A very small group of customers contributes significantly to revenue.

```

```

-- Checking for duplicate customer records before applying constraints

```

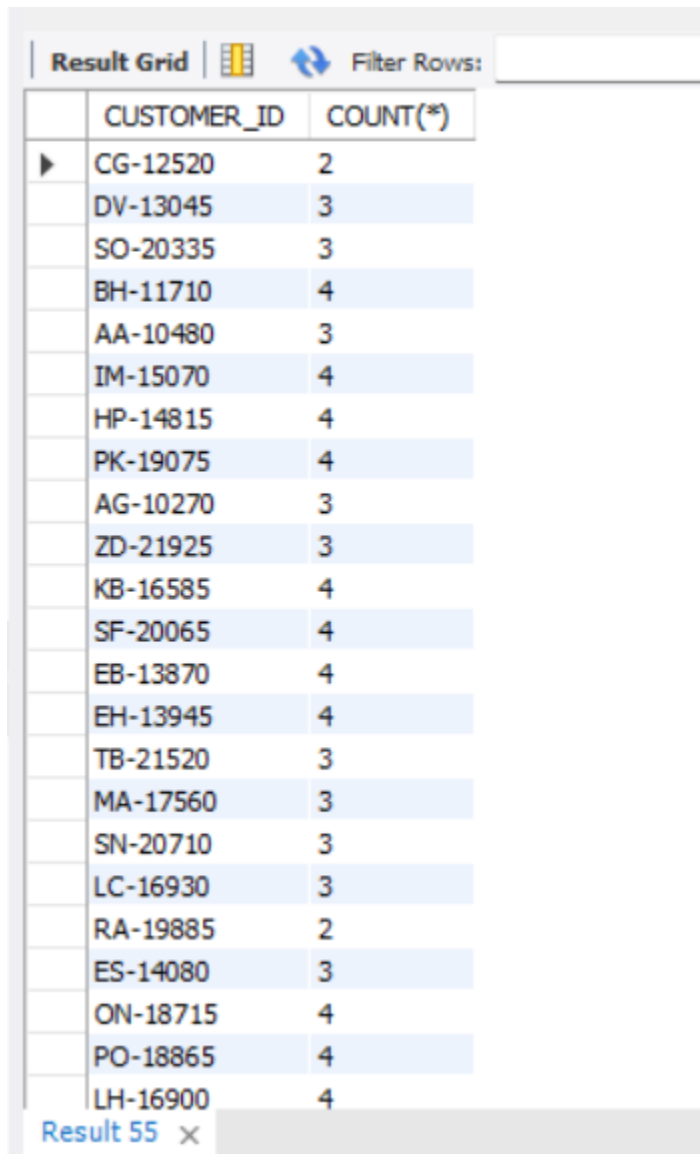
```

SELECT CUSTOMER_ID, COUNT(*)
FROM CUSTOMERS

```


GROUP BY CUSTOMER_ID

HAVING COUNT(*) > 1;



The screenshot shows a SQL query result grid with two columns: CUSTOMER_ID and COUNT(*). The grid contains 20 rows of data. The first row is highlighted with a blue arrow icon. The bottom of the grid shows 'Result 55' with a close button (X).

	CUSTOMER_ID	COUNT(*)
▶	CG-12520	2
	DV-13045	3
	SO-20335	3
	BH-11710	4
	AA-10480	3
	IM-15070	4
	HP-14815	4
	PK-19075	4
	AG-10270	3
	ZD-21925	3
	KB-16585	4
	SF-20065	4
	EB-13870	4
	EH-13945	4
	TB-21520	3
	MA-17560	3
	SN-20710	3
	LC-16930	3
	RA-19885	2
	ES-14080	3
	ON-18715	4
	PO-18865	4
	LH-16900	4

Result 55 X

-- This shows that our table contains duplicate values hence, a cleaned version is created for applying constraints

-- KEY CONSTRAINT :-

-- Creating cleaned tables by removing duplicates using GROUP BY to ensure uniqueness for primary key constraints

CREATE TABLE CUSTOMERS_CLEAN AS

SELECT

```
CUSTOMER_ID,  
MIN(CUSTOMER_NAME) AS CUSTOMER_NAME,  
MIN(SEGMENT) AS SEGMENT,  
MIN(REGION) AS REGION  
FROM CUSTOMERS  
GROUP BY CUSTOMER_ID;  
ALTER TABLE CUSTOMERS_CLEAN  
ADD PRIMARY KEY (CUSTOMER_ID);
```

```
CREATE TABLE PRODUCTS_CLEAN AS  
SELECT  
    PRODUCT_ID,  
    MIN(CATEGORY) AS CATEGORY,  
    MIN(SUB_CATEGORY) AS SUB_CATEGORY,  
    MIN(PRODUCT_NAME) AS PRODUCT_NAME  
FROM PRODUCTS  
GROUP BY PRODUCT_ID;  
ALTER TABLE PRODUCTS_CLEAN  
ADD PRIMARY KEY (PRODUCT_ID);
```

```
-- ORDER_ID is not unique in the dataset, so a surrogate key ROW_ID is created as primary key  
ALTER TABLE ORDERS  
ADD COLUMN ROW_ID INT AUTO_INCREMENT PRIMARY KEY;
```

```
-- Establishing relationships between tables using foreign keys to maintain referential integrity  
ALTER TABLE ORDERS
```

```
ADD CONSTRAINT FK_CUSTOMER  
FOREIGN KEY (CUSTOMER_ID)  
REFERENCES CUSTOMERS_CLEAN(CUSTOMER_ID);
```

```
ALTER TABLE ORDERS  
ADD CONSTRAINT FK_PRODUCT  
FOREIGN KEY (PRODUCT_ID)  
REFERENCES PRODUCTS_CLEAN(PRODUCT_ID);
```

/*Step 3: Final Combined Query

Write one final query that shows:

Customer Name

Total Sales

Rank

(Use JOIN + CTE + Window Function together)*/

-- PHASE 1 :- BUILD THE BASE JOIN (COMBINED ORDERS + CUSTOMERS)

```
SELECT  
    O.CUSTOMER_ID,  
    C.CUSTOMER_NAME,  
    O.SALES  
FROM ORDERS O  
JOIN CUSTOMERS_CLEAN C  
ON O.CUSTOMER_ID = C.CUSTOMER_ID;
```

Result Grid

 Filter Rows:

Export: 

Wrap Cell

	CUSTOMER_ID	CUSTOMER_NAME	SALES
▶	AA-10315	Alex Avila	362.94
	AA-10315	Alex Avila	11.54
	AA-10315	Alex Avila	26.96
	AA-10315	Alex Avila	673.568
	AA-10315	Alex Avila	52.98
	AA-10315	Alex Avila	3930.072
	AA-10315	Alex Avila	2.304
	AA-10315	Alex Avila	431.976
	AA-10315	Alex Avila	41.72
	AA-10315	Alex Avila	14.94
	AA-10315	Alex Avila	14.56
	AA-10375	Allen Arnold	16.768
	AA-10375	Allen Arnold	28.4
	AA-10375	Allen Arnold	149.97
	AA-10375	Allen Arnold	16.52
	AA-10375	Allen Arnold	5.248
	AA-10375	Allen Arnold	499.98
	AA-10375	Allen Arnold	5.28
	AA-10375	Allen Arnold	8.26
	AA-10375	Allen Arnold	34.272
	AA-10375	Allen Arnold	84.96
	AA-10375	Allen Arnold	14.952
	AA-10375	Allen Arnold	17.94

Result 56 ×

-- PHASE 2 :- CONVERT INTO CTE AND APPLY WINDOW FUNCTIONS

WITH CUSTOMER_TOTAL AS (

SELECT

O.CUSTOMER_ID,

C.CUSTOMER_NAME,

SUM(O.SALES) AS TOTAL_SALES

FROM ORDERS O

```
JOIN CUSTOMERS_CLEAN C
  ON O.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY O.CUSTOMER_ID, C.CUSTOMER_NAME
)
SELECT
  CUSTOMER_ID,
  CUSTOMER_NAME,
  TOTAL_SALES,
  RANK() OVER (ORDER BY TOTAL_SALES DESC) AS SALES_RANK,
  DENSE_RANK() OVER (ORDER BY TOTAL_SALES DESC) AS DENSERANK
FROM CUSTOMER_TOTAL;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	CUSTOMER_ID	CUSTOMER_NAME	TOTAL_SALES	SALES_RANK	DENSERANK
▶	SM-20320	Sean Miller	25043.05	1	1
	TC-20980	Tamara Chand	19017.847999999998	2	2
	RB-19360	Raymond Buch	15117.339	3	3
	TA-21385	Tom Ashbrook	14595.62	4	4
	AB-10105	Adrian Barton	14355.610999999997	5	5
	SC-20095	Sanjit Chand	14142.333999999999	6	6
	KL-16645	Ken Lonsdale	14071.917	7	7
	HL-15040	Hunter Lopez	12873.297999999999	8	8
	SE-20110	Sanjit Engle	12209.438	9	9
	CC-12370	Christopher Conant	12129.072	10	10
	TS-21370	Todd Sumrall	11885.871	11	11
	GT-14710	Greg Tran	11820.119999999997	12	12
	BM-11140	Becky Martin	11609.900000000001	13	13
	SV-20365	Seth Vernon	11420.645999999997	14	14
	CJ-12010	Caroline Jumper	11079.742	15	15
	CL-12565	Clay Ludtke	10880.545999999998	16	16
	ME-17320	Maria Etezadi	10635.917999999998	17	17
	KF-16285	Karen Ferguson	10604.266	18	18
	BS-11365	Bill Shonely	10501.652999999998	19	19
	EH-13765	Edward Hooks	10005.980000000001	20	20
	JL-15835	John Lee	9403.002999999999	21	21
	GT-14635	Grant Thornton	9351.211999999998	22	22
	HW-14935	Helen Wasserman	9300.253999999997	23	23

Result 57 ×

-- This query aggregates total sales per customer using a CTE, and ranks customers based on their sales performance using window functions (RANK and DENSE_RANK)

/*Mini Project: Customer Sales Insights

Answer the following using SQL:

Who are the top 5 customers?

Who are the bottom 5 customers?

Which customers made only one order?

Which customers have above-average sales?

What is the highest order value per customer? */

-- 1) Who are the top 5 customers?

```
WITH CUSTOMER_TOTAL AS (  
    SELECT  
        O.CUSTOMER_ID,  
        C.CUSTOMER_NAME,  
        SUM(O.SALES) AS TOTAL_SALES  
    FROM ORDERS O  
    JOIN CUSTOMERS_CLEAN C  
        ON O.CUSTOMER_ID = C.CUSTOMER_ID  
    GROUP BY O.CUSTOMER_ID, C.CUSTOMER_NAME  
)  
SELECT *  
FROM CUSTOMER_TOTAL  
ORDER BY TOTAL_SALES DESC  
LIMIT 5;
```

	CUSTOMER_ID	CUSTOMER_NAME	TOTAL_SALES
►	SM-20320	Sean Miller	25043.05
	TC-20980	Tamara Chand	19017.847999999998
	RB-19360	Raymond Buch	15117.339
	TA-21385	Tom Ashbrook	14595.62
	AB-10105	Adrian Barton	14355.610999999997

-- 2) Who are the bottom 5 customers?

```
WITH CUSTOMER_TOTAL AS (
```

```

SELECT
    O.CUSTOMER_ID,
    C.CUSTOMER_NAME,
    SUM(O.SALES) AS TOTAL_SALES
FROM ORDERS O
JOIN CUSTOMERS_CLEAN C
    ON O.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY O.CUSTOMER_ID, C.CUSTOMER_NAME
)
SELECT *
FROM CUSTOMER_TOTAL
ORDER BY TOTAL_SALES ASC
LIMIT 5;

```

Result Grid Filter Rows: Export: Wrap Cell Content:			
	CUSTOMER_ID	CUSTOMER_NAME	TOTAL_SALES
▶	TS-21085	Thais Sissman	4.833
	LD-16855	Lela Donovan	5.304
	MG-18205	Mitch Gastineau	12.32
	CJ-11875	Carl Jackson	16.52
	RS-19870	Roy Skaria	22.328000000000003

-- 3) Which customers made only one order?

```

SELECT
    O.CUSTOMER_ID,
    C.CUSTOMER_NAME,
    COUNT(O.ROW_ID) AS ORDER_COUNT

```

```

FROM ORDERS O
JOIN CUSTOMERS_CLEAN C
    ON O.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY O.CUSTOMER_ID, C.CUSTOMER_NAME
HAVING COUNT(O.ROW_ID) = 1;

```

Result Grid			
		Filter Rows:	
		Export:	
	CUSTOMER_ID	CUSTOMER_NAME	ORDER_COUNT
▶	AO-10810	Anthony O'Donnell	1
	CJ-11875	Carl Jackson	1
	JR-15700	Jocasta Rupert	1
	LD-16855	Lela Donovan	1
	MG-18205	Mitch Gastineau	1
	RE-19405	Ricardo Emerson	1

-- 4) Which customers have above-average sales?

```

WITH CUSTOMER_TOTAL AS (
    SELECT
        O.CUSTOMER_ID,
        C.CUSTOMER_NAME,
        SUM(O.SALES) AS TOTAL_SALES
    FROM ORDERS O
    JOIN CUSTOMERS_CLEAN C
        ON O.CUSTOMER_ID = C.CUSTOMER_ID
    GROUP BY O.CUSTOMER_ID, C.CUSTOMER_NAME
),
AVG_SALES AS (
    SELECT AVG(TOTAL_SALES) AS AVG_VALUE

```

```

FROM CUSTOMER_TOTAL
)
SELECT *
FROM CUSTOMER_TOTAL
WHERE TOTAL_SALES > (SELECT AVG_VALUE FROM AVG_SALES);

```

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
	CUSTOMER_ID	CUSTOMER_NAME	TOTAL_SALES
▶	AA-10315	Alex Avila	5563.56
	AA-10645	Anna Andreadi	4455.895
	AB-10060	Adam Bellavance	7732.669999999999
	AB-10105	Adrian Barton	14355.610999999997
	AC-10450	Amy Cox	5527.8460000000005
	AD-10180	Alan Dominguez	6100.999999999999
	AG-10675	Anna Gayman	3489.0395999999996
	AG-10900	Arthur Gainer	4510.7970000000005
	AH-10075	Adam Hart	3250.3369999999995
	AH-10210	Alan Hwang	4805.344
	AH-10690	Anna Häberlin	7888.294000000001
	AI-10855	Arianne Irving	4375.785999999999
	AJ-10780	Anthony Jacobs	4867.344
	AJ-10795	Anthony Johnson	4391.491
	AM-10705	Anne McFarland	5653.861999999999
	AP-10915	Arthur Pritchep	3323.556
	AR-10540	Andy Reiter	6608.448
	AR-10825	Anthony Rawles	4523.335999999999
	AS-10045	Aaron Smayling	3035.0320000000006
	AS-10090	Adam Shillingsburg	3255.309999999999
	AS-10225	Alan Schoenberger	4249
	AS-10285	Alejandro Savely	3212.272
	AT-10735	Annie Thurman	3831.8640000000005

-- 5) What is the highest order value per customer?

```

SELECT
O.CUSTOMER_ID,

```

```

C.CUSTOMER_NAME,
MAX(O.SALES) AS HIGHEST_ORDER_VALUE
FROM ORDERS O
JOIN CUSTOMERS_CLEAN C
ON O.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY O.CUSTOMER_ID, C.CUSTOMER_NAME;

```

Result Grid			
		Filter Rows:	Export: Wrap Cell Content:
	CUSTOMER_ID	CUSTOMER_NAME	HIGHEST_ORDER_VALUE
▶	AA-10315	Alex Avila	3930.072
	AA-10375	Allen Arnold	499.98
	AA-10480	Andrew Allen	479.97
	AA-10645	Anna Andreadi	1323.9
	AB-10015	Aaron Bergman	341.96
	AB-10060	Adam Bellavance	4355.168
	AB-10105	Adrian Barton	9892.74
	AB-10150	Aimee Bixby	479.97
	AB-10165	Alan Barnes	304.23
	AB-10255	Alejandro Ballentine	479.984
	AB-10600	Ann Blume	899.136
	AC-10420	Alyssa Crouse	698.352
	AC-10450	Amy Cox	1793.98
	AC-10615	Ann Chong	1282.41
	AC-10660	Anna Chung	201.584
	AD-10180	Alan Dominguez	2999.95
	AF-10870	Art Ferguson	662.84
	AF-10885	Art Foster	349.965
	AG-10270	Alejandro Grove	1336.44
	AG-10300	Aleksandra Ganna...	230.28
	AG-10330	Alex Grayson	182.55
	AG-10390	Allen Goldenen	71.28
	AG-10495	Andrew Giertsen	516.96

Result 62 ×

-- =====

-- FINAL SUMMARY:

-- This project demonstrates end to end SQL skills including: data cleaning, normalization, constraint handling, and advanced analytical queries to extract business insights.

-- Key Findings:

-- A small group of customers contributes most revenue

-- Some customers show only one time purchase behavior

-- High value customers can be identified using ranking

-- This analysis can help businesses improve customer targeting and revenue optimization strategies.

-- =====